

Quasi 0-D/1-D Fontan Model Technical Reference

Standardized model definition, equations, segments, and free parameters

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1 Quasi 0-D/1-D Fontan Model Technical Reference

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1.1 Model Construction

1.1.1 Scope and Status

Accepted PhysioBlocks-only quasi 0-D/1-D model.

This model keeps the accepted full 0-D heart, atrium, valves, systemic beds, pulmonary RCR beds, and fenestration, while replacing selected aortic and Fontan conduit shortcuts with distributed R-L-C chains.

It does not contain a nonlinear 1-D PDE solver. Its quasi-1-D behavior comes from repeated hydraulic R-L links and compliance states embedded in the closed loop.

The accepted baseline is selected by the frozen quasi-vs-full0D superiority gate.

Quasi 0-D/1-D Fontan schematic

Same closed-loop components as full 0-D, with aortic and Fontan pathways replaced by PhysioBlocks R-L-C chains. Nominal flow follows the arrows.

Accepted quasi config: corrected AAo/arch x4 and DAo x6, SVC x3, IVC x5, RPA x3, LPA x4, proximal pulmonary split and prebail restoration included.

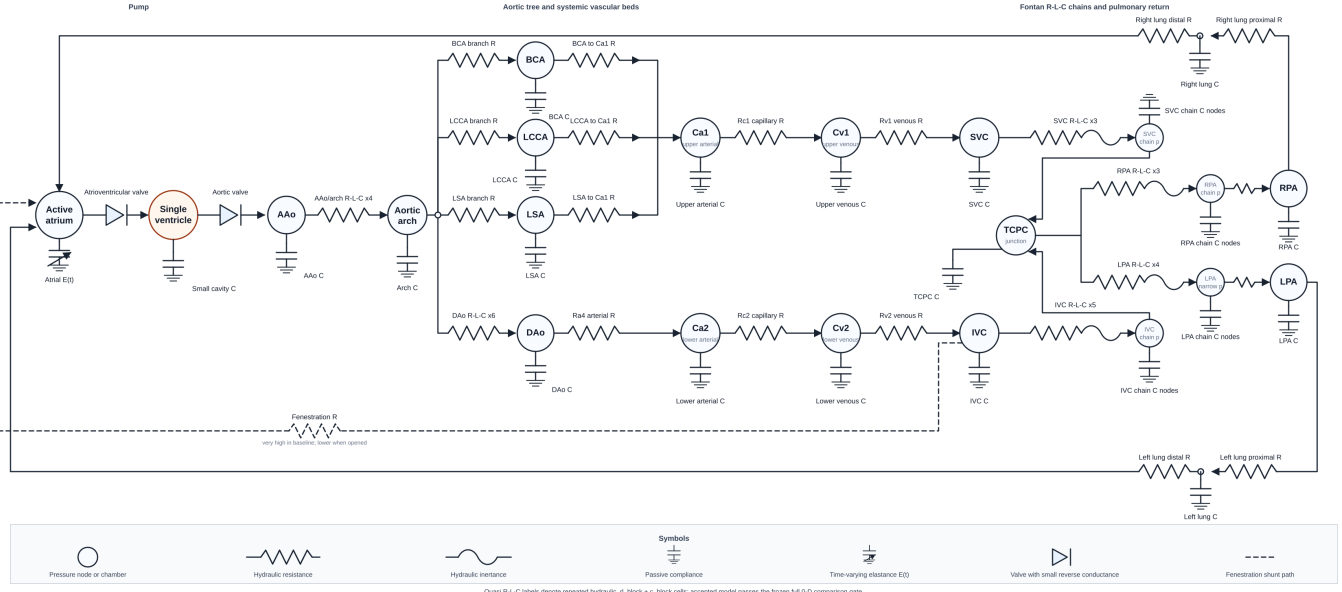


Figure 1: quasi_0d_1d schematic

1.1.2 Schematic

1.1.3 Accepted Components

- active atrium and active spherical ventricle retained from full 0-D
- atrioventricular and aortic valve R-L blocks retained from full 0-D
- AAo/arch, DAo, SVC, IVC, RPA, and LPA quasi R-L-C chains
- upper and lower systemic vascular beds
- right and left pulmonary RCR Windkessel beds
- scenario-specific pulmonary vasodilation, fenestration, and LPA obstruction variants without scenario-specific retuning

1.1.4 Authoritative Baseline Config

The executable topology and free-parameter values are taken from `models/quasi_0d_1d/configs/fontan_quasi_baseline.jsonc`

- pressure nodes: 36
- blocks/segments: 84
- free parameter entries: 146
- boundary conditions: 0

1.1.5 Scenario Configs

- `models/quasi_0d_1d/configs/fontan_quasi_baseline.jsonc`
- `models/quasi_0d_1d/configs/fontan_quasi_fenestration.jsonc`
- `models/quasi_0d_1d/configs/fontan_quasi_lpa_obstruction.jsonc`
- `models/quasi_0d_1d/configs/fontan_quasi_smoke.jsonc`
- `models/quasi_0d_1d/configs/fontan_quasi_vasodilation.jsonc`

1.1.6 Pressure Nodes

- atrial
- cavity
- aao
- aortic_arch
- bca

- lcca
- lsa
- upper_art
- upper_ven
- dao
- lower_art
- lower_ven
- svc
- ivc
- tcpc
- rpa
- lpa
- quasi_aao_arch_p_01
- quasi_aao_arch_p_02
- quasi_aao_arch_p_03
- quasi_dao_p_01
- quasi_dao_p_02
- quasi_dao_p_03
- quasi_dao_p_04
- quasi_dao_p_05
- quasi_svc_p_01
- quasi_svc_p_02
- quasi_ivc_p_01
- quasi_ivc_p_02
- quasi_ivc_p_03
- quasi_ivc_p_04
- quasi_rpa_p_01
- quasi_rpa_p_02
- quasi_lpa_p_01
- quasi_lpa_p_02
- quasi_lpa_p_03

1.1.7 Block Type Counts

Block type	Count
c_block	41
hydraulic_rl_block	25
rc_block	12
rcr_block	2
spherical_cavity_block	1
time_varying_elastance_atrium_block	1
valve_rl_block	2

1.2 Governing Equations

The sign convention follows PhysioBlocks local block fluxes. For a two-node flow element, local node 1 is the first node listed in the config and local node 2 is the second node listed in the config.

1.2.1 Nodal Conservation

At every pressure node, the algebraic/differential network residual is the sum of all block flux contributions attached to that node:

$$\sum_{b \in \mathcal{B}(i)} Q_{b,i} = 0.$$

Storage blocks contribute pressure derivatives to this same residual, so closed-loop volume conservation is enforced through the connected block equations rather than through prescribed boundary flow.

1.2.2 Passive Compliance Block

For a `c_block` at pressure node `P` with capacitance `C`:

$$Q = -C \frac{dP}{dt}.$$

The saved stored volume is proportional to pressure in the local linear compliance approximation:

$$V = CP.$$

1.2.3 Pure RC Resistor Convention

This repository uses `rc_block` with zero capacitance as a pure resistive link. `PhysioBlocks` defines the local fluxes as:

$$Q_1 = \frac{P_2 - P_1}{R},$$
$$Q_2 = \frac{P_1 - P_2}{R} - C \frac{dP_2}{dt}.$$

When `C` = 0, the block is used as a pure resistor. To represent an upstream-to-downstream path with positive physical flow from upstream to downstream, the configs assign local node 2 to the upstream pressure and local node 1 to the downstream pressure.

1.2.4 Hydraulic R-L Link

The local quasi-vessel R-L element uses positive internal flow from local node 1 to local node 2:

$$L \frac{dQ}{dt} + RQ = P_1 - P_2,$$
$$Q_1 = -Q, \quad Q_2 = Q.$$

This is the repeated segment equation used by the quasi 0-D/1-D chains.

1.2.5 Valve R-L Block

The R-L valve block has local positive flow from node 1 to node 2 and switches conductance according to flow direction:

$$L \frac{dQ}{dt} + P_2 - P_1 + R(Q)Q = 0,$$
$$R(Q) = \begin{cases} 1/G_f, & Q > 0, \\ 1/G_b, & Q < 0, \end{cases}$$
$$Q_1 = -Q, \quad Q_2 = Q.$$

`G_f` is the forward conductance and `G_b` is the backward conductance.

1.2.6 Pulmonary RCR Windkessel

For an `rcr_block` with inlet pressure P_1 , outlet pressure P_2 , middle pressure P_m , proximal resistance R_1 , distal resistance R_2 , and compliance C :

$$Q_1 = \frac{P_m - P_1}{R_1},$$

$$Q_2 = \frac{P_m - P_2}{R_2},$$

$$\frac{P_1 - P_m}{R_1} + \frac{P_2 - P_m}{R_2} - C \frac{dP_m}{dt} = 0.$$

1.2.7 Active Atrium

The active atrium is a one-node time-varying elastance chamber:

$$E(t) = E_{min} + (E_{max} - E_{min})a(t),$$

$$V_a(t) = V_{0,a} + \frac{P_a(t) - P_{ext}}{E(t)},$$

$$Q_a = -\frac{dV_a}{dt}.$$

The activation $a(t)$ is a raised-cosine pulse over the configured start, peak, and end phase of the cardiac cycle.

1.2.8 Spherical Ventricular Cavity

The active ventricular cavity stores volume through the spherical cavity displacement y , reference radius R_0 , and wall thickness d_0 :

$$V(y) = \frac{4\pi}{3} \left[R_0 \left(1 + \frac{y}{R_0} \right) - \frac{d_0}{2 \left(1 + \frac{y}{R_0} \right)^2} \right]^3,$$

$$Q_v = -\frac{dV(y)}{dt}.$$

The pressure-displacement relation comes from the configured PhysioBlocks spherical dynamics, velocity law, passive rheology, and active macro-Huxley submodels. The corresponding free parameters are listed in the parameter inventory below with the `cavity.*` prefix.

1.3 Quasi Chain Summary

The accepted quasi chains are assembled from repeated hydraulic R-L links and downstream compliance states. Total values in this table are computed from the baseline config.

Chain	Path	Segments	Total R (Pa s m ⁻³)	Total L (Pa s ² m ⁻³)	Total C (m ³ Pa ⁻¹)
aao_arch	aao -> aortic_arch	4	207755	100223	3.14191e-10
dao	aortic_arch -> dao	6	3.56544e+06	744674	3.9886e-10
svc	svc -> tcpc	3	5.86617e+06	62815.8	4.76865e-10
ivc	ivc -> tcpc	5	4.39963e+06	108574	3.66194e-09

Chain	Path	Segments	Total R (Pa s m^{-3})	Total L ($\text{Pa s}^2 \text{m}^{-3}$)	Total C ($\text{m}^3 \text{Pa}^{-1}$)
rpa	tcpc -> rpa	3	2.3998e+06	95832	1.903e-10
lpa	tcpc -> lpa	4	3.46637e+06	841348	2.10762e-10

1.4 Segment Inventory

Each row is a block or segment in the accepted baseline config. The parameter column lists explicit block fields and all config parameters sharing the block-name prefix.

Segment/block	Type	Local nodes	Free-parameter fields
cavity	spherical_cavity_block	1: cavity	disp = cavity.dynamics.disp; radius -> heart_radius; thickness -> heart_thickness; cav- ity.dynamics.damping_coef; cav- ity.dynamics.hyperelastic_cst; cavity.dynamics.vol_mass; cav- ity.rheology.active_law.activation; cav- ity.rheology.active_law.crossbridge_sti cav- ity.rheology.active_law.destruction_ra cav- ity.rheology.active_law.starling_abscis cav- ity.rheology.active_law.starling_ordin cav- ity.rheology.damping_parallel; cav- ity.rheology.series_stiffness; cav- ity.velocity_law.scheme_ts_hht backward_conductance -> valves.backward_conductance; valve_atrium.conductance; valve_atrium.inductance; valve_atrium.scheme_ts_flux backward_conductance -> valves.backward_conductance; valve_arterial.conductance; valve_arterial.inductance; valve_arterial.scheme_ts_flux capacitance_valve.capacitance
valve_atrium	valve_rl_block	1: atrial; 2: cavity	
valve_arterial	valve_rl_block	1: cavity; 2: aao	
capacitance_valve	c_block	1: cavity	

Segment/block	Type	Local nodes	Free-parameter fields
active_atrium	time_varying_elastance_atrium_block		pressure_external -> pleural.pressure; elastance_min -> ac- tive_atrium.elastance_min; elastance_max -> ac- tive_atrium.elastance_max; unstressed_volume -> ac- tive_atrium.unstressed_volume; activation_start -> ac- tive_atrium.activation_start; activation_peak -> ac- tive_atrium.activation_peak; activation_end -> ac- tive_atrium.activation_end; heartbeat_duration -> heartbeat_duration
aao_compliance	c_block	1: aao	aao_compliance.capacitance
aortic_arch_compliance	c_block	1: aortic_arch	aortic_arch_compliance.capacitance
bca_compliance	c_block	1: bca	bca_compliance.capacitance
lcca_compliance	c_block	1: lcca	lcca_compliance.capacitance
lsa_compliance	c_block	1: lsa	lsa_compliance.capacitance
upper_ca1	c_block	1: upper_art	upper_ca1.capacitance
upper_cv1	c_block	1: upper_ven	upper_cv1.capacitance
dao_compliance	c_block	1: dao	dao_compliance.capacitance
lower_ca2	c_block	1: lower_art	lower_ca2.capacitance
lower_cv2	c_block	1: lower_ven	lower_cv2.capacitance
svc_compliance	c_block	1: svc	svc_compliance.capacitance
ivc_compliance	c_block	1: ivc	ivc_compliance.capacitance
tcpc_compliance	c_block	1: tcpc	tcpc_compliance.capacitance
rpa_compliance	c_block	1: rpa	rpa_compliance.capacitance
lpa_compliance	c_block	1: lpa	lpa_compliance.capacitance
arch_bca	rc_block	1: bca; 2: aortic_arch	capacitance -> zero_capacitance; arch_bca.resistance
upper_bca_to_ca1	rc_block	1: upper_art; 2: bca	capacitance -> zero_capacitance; up- per_bca_to_ca1.resistance
arch_lcca	rc_block	1: lcca; 2: aortic_arch	capacitance -> zero_capacitance; arch_lcca.resistance
upper_lcca_to_ca1	rc_block	1: upper_art; 2: lcca	capacitance -> zero_capacitance; up- per_lcca_to_ca1.resistance
arch_lsa	rc_block	1: lsa; 2: aortic_arch	capacitance -> zero_capacitance; arch_lsa.resistance
upper_lsa_to_ca1	rc_block	1: upper_art; 2: lsa	capacitance -> zero_capacitance; up- per_lsa_to_ca1.resistance
upper_rc1	rc_block	1: upper_ven; 2: upper_art	capacitance -> zero_capacitance; upper_rc1.resistance
upper_rv1	rc_block	1: svc; 2: upper_ven	capacitance -> zero_capacitance; upper_rv1.resistance

Segment/block	Type	Local nodes	Free-parameter fields
lower_ra4	rc_block	1: lower_art; 2: dao	capacitance -> zero_capacitance; lower_ra4.resistance
lower_rc2	rc_block	1: lower_ven; 2: lower_art	capacitance -> zero_capacitance; lower_rc2.resistance
lower_rv2	rc_block	1: ivc; 2: lower_ven	capacitance -> zero_capacitance; lower_rv2.resistance
right_lung	rcr_block	1: rpa; 2: atrial	pressure_mid = right_lung.pressure_mid; resistance_1 -> right_lung.resistance_1; resistance_2 -> right_lung.resistance_2; capacitance -> right_lung.capacitance
left_lung	rcr_block	1: lpa; 2: atrial	pressure_mid = left_lung.pressure_mid; resistance_1 -> left_lung.resistance_1; resistance_2 -> left_lung.resistance_2; capacitance -> left_lung.capacitance
fenestration	rc_block	1: atrial; 2: ivc	capacitance -> zero_capacitance; fenestration.resistance
quasi_aao_arch_rl_01	hydraulic_rl_block	1: aao; 2: quasi_aao_arch_p_01	resistance -> quasi_aao_arch_rl_01.resistance; inductance -> quasi_aao_arch_rl_01.inductance
quasi_aao_arch_c_01	c_block	1: quasi_aao_arch_p_01	quasi_aao_arch_c_01.capacitance
quasi_aao_arch_rl_02	hydraulic_rl_block	1: quasi_aao_arch_p_01; 2: quasi_aao_arch_p_02	resistance -> quasi_aao_arch_rl_02.resistance; inductance -> quasi_aao_arch_rl_02.inductance
quasi_aao_arch_c_02	c_block	1: quasi_aao_arch_p_02	quasi_aao_arch_c_02.capacitance
quasi_aao_arch_rl_03	hydraulic_rl_block	1: quasi_aao_arch_p_02; 2: quasi_aao_arch_p_03	resistance -> quasi_aao_arch_rl_03.resistance; inductance -> quasi_aao_arch_rl_03.inductance
quasi_aao_arch_c_03	c_block	1: quasi_aao_arch_p_03	quasi_aao_arch_c_03.capacitance
quasi_aao_arch_rl_04	hydraulic_rl_block	1: quasi_aao_arch_p_03; 2: aortic_arch	resistance -> quasi_aao_arch_rl_04.resistance; inductance -> quasi_aao_arch_rl_04.inductance
quasi_aao_arch_c_04	c_block	1: aortic_arch	quasi_aao_arch_c_04.capacitance
quasi_dao_rl_01	hydraulic_rl_block	1: aortic_arch; 2: quasi_dao_p_01	resistance -> quasi_dao_rl_01.resistance; inductance -> quasi_dao_rl_01.inductance
quasi_dao_c_01	c_block	1: quasi_dao_p_01	quasi_dao_c_01.capacitance
quasi_dao_rl_02	hydraulic_rl_block	1: quasi_dao_p_01; 2: quasi_dao_p_02	resistance -> quasi_dao_rl_02.resistance; inductance -> quasi_dao_rl_02.inductance

Segment/block	Type	Local nodes	Free-parameter fields
quasi_dao_c_02 quasi_dao_rl_03	c_block hydraulic_rl_block	1: quasi_dao_p_02 1: quasi_dao_p_02; 2: quasi_dao_p_03	quasi_dao_c_02.capacitance resistance -> quasi_dao_rl_03.resistance; inductance -> quasi_dao_rl_03.inductance
quasi_dao_c_03 quasi_dao_rl_04	c_block hydraulic_rl_block	1: quasi_dao_p_03 1: quasi_dao_p_03; 2: quasi_dao_p_04	quasi_dao_c_03.capacitance resistance -> quasi_dao_rl_04.resistance; inductance -> quasi_dao_rl_04.inductance
quasi_dao_c_04 quasi_dao_rl_05	c_block hydraulic_rl_block	1: quasi_dao_p_04 1: quasi_dao_p_04; 2: quasi_dao_p_05	quasi_dao_c_04.capacitance resistance -> quasi_dao_rl_05.resistance; inductance -> quasi_dao_rl_05.inductance
quasi_dao_c_05 quasi_dao_rl_06	c_block hydraulic_rl_block	1: quasi_dao_p_05 1: quasi_dao_p_05; 2: dao	quasi_dao_c_05.capacitance resistance -> quasi_dao_rl_06.resistance; inductance -> quasi_dao_rl_06.inductance
quasi_dao_c_06 quasi_svc_rl_01	c_block hydraulic_rl_block	1: dao 1: svc; 2: quasi_svc_p_01	quasi_dao_c_06.capacitance resistance -> quasi_svc_rl_01.resistance; inductance -> quasi_svc_rl_01.inductance
quasi_svc_c_01 quasi_svc_rl_02	c_block hydraulic_rl_block	1: quasi_svc_p_01 1: quasi_svc_p_01; 2: quasi_svc_p_02	quasi_svc_c_01.capacitance resistance -> quasi_svc_rl_02.resistance; inductance -> quasi_svc_rl_02.inductance
quasi_svc_c_02 quasi_svc_rl_03	c_block hydraulic_rl_block	1: quasi_svc_p_02 1: quasi_svc_p_02; 2: tcpc	quasi_svc_c_02.capacitance resistance -> quasi_svc_rl_03.resistance; inductance -> quasi_svc_rl_03.inductance
quasi_svc_c_03 quasi_ivc_rl_01	c_block hydraulic_rl_block	1: tcpc 1: ivc; 2: quasi_ivc_p_01	quasi_svc_c_03.capacitance resistance -> quasi_ivc_rl_01.resistance; inductance -> quasi_ivc_rl_01.inductance
quasi_ivc_c_01 quasi_ivc_rl_02	c_block hydraulic_rl_block	1: quasi_ivc_p_01 1: quasi_ivc_p_01; 2: quasi_ivc_p_02	quasi_ivc_c_01.capacitance resistance -> quasi_ivc_rl_02.resistance; inductance -> quasi_ivc_rl_02.inductance
quasi_ivc_c_02 quasi_ivc_rl_03	c_block hydraulic_rl_block	1: quasi_ivc_p_02 1: quasi_ivc_p_02; 2: quasi_ivc_p_03	quasi_ivc_c_02.capacitance resistance -> quasi_ivc_rl_03.resistance; inductance -> quasi_ivc_rl_03.inductance
quasi_ivc_c_03 quasi_ivc_rl_04	c_block hydraulic_rl_block	1: quasi_ivc_p_03 1: quasi_ivc_p_03; 2: quasi_ivc_p_04	quasi_ivc_c_03.capacitance resistance -> quasi_ivc_rl_04.resistance; inductance -> quasi_ivc_rl_04.inductance
quasi_ivc_c_04	c_block	1: quasi_ivc_p_04	quasi_ivc_c_04.capacitance

Segment/block	Type	Local nodes	Free-parameter fields
quasi_ivc_rl_05	hydraulic_rl_block	1: quasi_ivc_p_04; 2: tcpc	resistance -> quasi_ivc_rl_05.resistance; inductance -> quasi_ivc_rl_05.inductance
quasi_ivc_c_05	c_block	1: tcpc	quasi_ivc_c_05.capacitance
quasi_rpa_rl_01	hydraulic_rl_block	1: tcpc; 2: quasi_rpa_p_01	resistance -> quasi_rpa_rl_01.resistance; inductance -> quasi_rpa_rl_01.inductance
quasi_rpa_c_01	c_block	1: quasi_rpa_p_01	quasi_rpa_c_01.capacitance
quasi_rpa_rl_02	hydraulic_rl_block	1: quasi_rpa_p_01; 2: quasi_rpa_p_02	resistance -> quasi_rpa_rl_02.resistance; inductance -> quasi_rpa_rl_02.inductance
quasi_rpa_c_02	c_block	1: quasi_rpa_p_02	quasi_rpa_c_02.capacitance
quasi_rpa_rl_03	hydraulic_rl_block	1: quasi_rpa_p_02; 2: rpa	resistance -> quasi_rpa_rl_03.resistance; inductance -> quasi_rpa_rl_03.inductance
quasi_rpa_c_03	c_block	1: rpa	quasi_rpa_c_03.capacitance
quasi_lpa_rl_01	hydraulic_rl_block	1: tcpc; 2: quasi_lpa_p_01	resistance -> quasi_lpa_rl_01.resistance; inductance -> quasi_lpa_rl_01.inductance
quasi_lpa_c_01	c_block	1: quasi_lpa_p_01	quasi_lpa_c_01.capacitance
quasi_lpa_rl_02	hydraulic_rl_block	1: quasi_lpa_p_01; 2: quasi_lpa_p_02	resistance -> quasi_lpa_rl_02.resistance; inductance -> quasi_lpa_rl_02.inductance
quasi_lpa_c_02	c_block	1: quasi_lpa_p_02	quasi_lpa_c_02.capacitance
quasi_lpa_rl_03	hydraulic_rl_block	1: quasi_lpa_p_02; 2: quasi_lpa_p_03	resistance -> quasi_lpa_rl_03.resistance; inductance -> quasi_lpa_rl_03.inductance
quasi_lpa_c_03	c_block	1: quasi_lpa_p_03	quasi_lpa_c_03.capacitance
quasi_lpa_rl_04	hydraulic_rl_block	1: quasi_lpa_p_03; 2: lpa	resistance -> quasi_lpa_rl_04.resistance; inductance -> quasi_lpa_rl_04.inductance
quasi_lpa_c_04	c_block	1: lpa	quasi_lpa_c_04.capacitance

1.5 Free Parameters

Unless a parameter states otherwise in the config, units follow the repository SI convention: pressure in Pa, flow in m^3s^{-1} , resistance in Pa s m^{-3} , capacitance in m^3Pa^{-1} , inertance in $\text{Pa s}^2\text{m}^{-3}$, volume in m^3 , length in m, and time in s.

The entries below are the complete `parameters` dictionary from the authoritative baseline config. Derived-expression entries are shown as compact JSON.

- `aao_compliance.capacitance = 8.0006800578e-11`
- `active_atrium.activation_end = 0.98`
- `active_atrium.activation_peak = 0.9`
- `active_atrium.activation_start = 0.78`
- `active_atrium.elastance_max = 66661000`
- `active_atrium.elastance_min = 16665250`
- `active_atrium.unstressed_volume = 4.2e-05`

- active_law.activation.max = 35
- active_law.activation.min = -20
- aortic_arch_compliance.capacitance = 1.00008500723e-10
- arch_bca.resistance = 5332880
- arch_lcca.resistance = 7999320
- arch_lsa.resistance = 7999320
- bca_compliance.capacitance = 3.00025502168e-09
- capacitance_valve.capacitance = 5e-12
- cavity.dynamics.damping_coef = 70
- cavity.dynamics.hyperelastic_cst = [444.0,2.9,69.0,6.5]
- cavity.dynamics.vol_mass = 1000
- cavity.rheology.active_law.activation = {"alpha": "diastole_scaling_factor", "phases": [0,0,1,1,1,0,0], "ref"
- cavity.rheology.active_law.crossbridge_stiffness = 273000
- cavity.rheology.active_law.destruction_rate = 12
- cavity.rheology.active_law.starling_abscissas = [-0.1668,-0.0073,0.0534,0.0969,0.1326,0.2016,0.4663,0.91
- cavity.rheology.active_law.starling_ordinates = [0.0,0.5614,0.7748,0.8933,0.9618,1.0,1.0,0.1075,0.0]
- cavity.rheology.damping_parallel = 70
- cavity.rheology.series_stiffness = 100000000
- cavity.velocity_law.scheme_ts_hht = 0.4
- conduit.scheme_ts_flux = 0.25
- dao_compliance.capacitance = 1.80015301301e-10
- diastole_scaling_factor = 0.8
- fenestration.resistance = 1.33322e+14
- heart_contractility = 39690
- heart_radius = 0.0245531
- heart_rate = 69.9300699301
- heart_thickness = 0.008160295
- heartbeat_duration = {"factors": [60.0], "inverses": ["heart_rate"], "type": "product"}
- ivc_compliance.capacitance = 2.13768170294e-07
- lcca_compliance.capacitance = 1.50012751084e-09
- left_lung.capacitance = 3.00025502168e-08
- left_lung.resistance_1 = 15546678.42
- left_lung.resistance_2 = 8371288.38
- lower_ca2.capacitance = 1.25010625903e-08
- lower_cv2.capacitance = 8.55072681178e-08
- lower_ra4.resistance = 229804464.96
- lower_rc2.resistance = 9827164.62
- lower_rv2.resistance = 2267807.22
- lpa_compliance.capacitance = 1.12509563313e-08
- lsa_compliance.capacitance = 1.50012751084e-09
- pleural.pressure = 0
- quasi_aao_arch_c_01.capacitance = 1.06684947589e-10
- quasi_aao_arch_c_02.capacitance = 8.65153768159e-11
- quasi_aao_arch_c_03.capacitance = 6.84630538033e-11
- quasi_aao_arch_c_04.capacitance = 5.25279785514e-11
- quasi_aao_arch_rl_01.inductance = 17213.346097
- quasi_aao_arch_rl_01.resistance = 22695.9439317
- quasi_aao_arch_rl_02.inductance = 21226.3414179
- quasi_aao_arch_rl_02.resistance = 34619.085625
- quasi_aao_arch_rl_03.inductance = 26823.298468
- quasi_aao_arch_rl_03.resistance = 55522.8933759
- quasi_aao_arch_rl_04.inductance = 34960.5101288
- quasi_aao_arch_rl_04.resistance = 94917.4266251
- quasi_dao_c_01.capacitance = 8.84084249797e-11
- quasi_dao_c_02.capacitance = 7.89166401957e-11
- quasi_dao_c_03.capacitance = 6.99641613653e-11
- quasi_dao_c_04.capacitance = 6.15509884886e-11
- quasi_dao_c_05.capacitance = 5.36771215655e-11

- quasi_dao_c_06.capacitance = 4.6342560596e-11
- quasi_dao_rl_01.inductance = 88916.2832192
- quasi_dao_rl_01.resistance = 290023.503509
- quasi_dao_rl_02.inductance = 99610.7859504
- quasi_dao_rl_02.resistance = 364163.26054
- quasi_dao_rl_03.inductance = 112356.789549
- quasi_dao_rl_03.resistance = 463591.768637
- quasi_dao_rl_04.inductance = 127714.416088
- quasi_dao_rl_04.resistance = 599408.164486
- quasi_dao_rl_05.inductance = 146448.772311
- quasi_dao_rl_05.resistance = 788837.155596
- quasi_dao_rl_06.inductance = 169626.979031
- quasi_dao_rl_06.resistance = 1059418.36317
- quasi_ivc_c_01.capacitance = 9.36020049527e-10
- quasi_ivc_c_02.capacitance = 8.27515133881e-10
- quasi_ivc_c_03.capacitance = 7.25698877418e-10
- quasi_ivc_c_04.capacitance = 6.30571280139e-10
- quasi_ivc_c_05.capacitance = 5.42132342044e-10
- quasi_ivc_rl_01.inductance = 16371.4663236
- quasi_ivc_rl_01.resistance = 481040.236632
- quasi_ivc_rl_02.inductance = 18518.1153694
- quasi_ivc_rl_02.resistance = 615845.339818
- quasi_ivc_rl_03.inductance = 21116.2249191
- quasi_ivc_rl_03.resistance = 801382.312152
- quasi_ivc_rl_04.inductance = 24301.8056828
- quasi_ivc_rl_04.resistance = 1062399.70537
- quasi_ivc_rl_05.inductance = 28266.1991006
- quasi_ivc_rl_05.resistance = 1438958.40603
- quasi_lpa.narrowing_radius_m = 0.003
- quasi_lpa.narrowing_resistance_scale = 1
- quasi_lpa_c_01.capacitance = 9.20271250531e-11
- quasi_lpa_c_02.capacitance = 5.37066818133e-11
- quasi_lpa_c_03.capacitance = 2.79163314694e-11
- quasi_lpa_c_04.capacitance = 3.71122994829e-11
- quasi_lpa_rl_01.inductance = 166516.34733
- quasi_lpa_rl_01.resistance = 418377.82423
- quasi_lpa_rl_02.inductance = 285328.01137
- quasi_lpa_rl_02.resistance = 1299644.4096
- quasi_lpa_rl_03.inductance = 222292.645222
- quasi_lpa_rl_03.resistance = 1119737.36501
- quasi_lpa_rl_04.inductance = 167211.281804
- quasi_lpa_rl_04.resistance = 628612.401151
- quasi_rpa_c_01.capacitance = 6.34334120112e-11
- quasi_rpa_c_02.capacitance = 6.34334120112e-11
- quasi_rpa_c_03.capacitance = 6.34334120112e-11
- quasi_rpa_rl_01.inductance = 31943.9980454
- quasi_rpa_rl_01.resistance = 799932
- quasi_rpa_rl_02.inductance = 31943.9980454
- quasi_rpa_rl_02.resistance = 799932
- quasi_rpa_rl_03.inductance = 31943.9980454
- quasi_rpa_rl_03.resistance = 799932
- quasi_svc_c_01.capacitance = 2.02745289685e-10
- quasi_svc_c_02.capacitance = 1.57000040638e-10
- quasi_svc_c_03.capacitance = 1.1711956711e-10
- quasi_svc_rl_01.inductance = 15616.2443439
- quasi_svc_rl_01.resistance = 1026122.68543
- quasi_svc_rl_02.inductance = 20166.3641004
- quasi_svc_rl_02.resistance = 1720931.87678

- `quasi_svc_rl_03.inductance = 27033.2281907`
- `quasi_svc_rl_03.resistance = 3119113.43779`
- `right_lung.capacitance = 3.00025502168e-08`
- `right_lung.resistance_1 = 10763085.06`
- `right_lung.resistance_2 = 5795507.34`
- `rpa_compliance.capacitance = 1.12509563313e-08`
- `svc_compliance.capacitance = 1.4251211353e-07`
- `tcpc_compliance.capacitance = 1.20010200867e-09`
- `upper_bca_to_ca1.resistance = 49595784`
- `upper_ca1.capacitance = 1.50012751084e-08`
- `upper_cv1.capacitance = 5.70048454119e-08`
- `upper_lcca_to_ca1.resistance = 99458212`
- `upper_lsa_to_ca1.resistance = 99458212`
- `upper_rc1.resistance = 156439020.571`
- `upper_rv1.resistance = 67045294.5306`
- `valve_arterial.conductance = 1.3e-05`
- `valve_arterial.inductance = 30000`
- `valve_arterial.scheme_ts_flux = 0.25`
- `valve_atrium.conductance = 9e-06`
- `valve_atrium.inductance = 1000`
- `valve_atrium.scheme_ts_flux = 0.25`
- `valves.backward_conductance = 5e-12`
- `zero_capacitance = 0`

1.6 Documentation and Regeneration

Model-local documentation artifacts:

- `models/quasi_0d_1d/README.md`
- `models/quasi_0d_1d/docs/quasi_0d_1d_schematic.svg`
- `models/quasi_0d_1d/docs/quasi_0d_1d_schematic.png`
- `models/quasi_0d_1d/docs/implementation_notes.md`
- `models/quasi_0d_1d/docs/quasi_0d_1d_technical_reference.md`
- `models/quasi_0d_1d/docs/quasi_0d_1d_technical_reference.pdf`

Regenerate the technical reference source and PDF with:

```
python3 scripts/docs/build_model_reference_pdfs.py --model quasi_0d_1d
```

1.7 Current Limitations

- The model is quasi 0-D/1-D only; wave propagation is approximated by finite R-L-C chains rather than by a true coupled 1-D solver.
- Clinical D_{AO} bed-entry flow remains a soft diagnostic because it mixes aortic trunk behavior with downstream terminal-load dynamics.
- The model is not clinically validated.

The model parameters and standardized data are for computational development and calibration workflows. Simulation outputs must not be presented as clinically validated without separate validation and documentation.